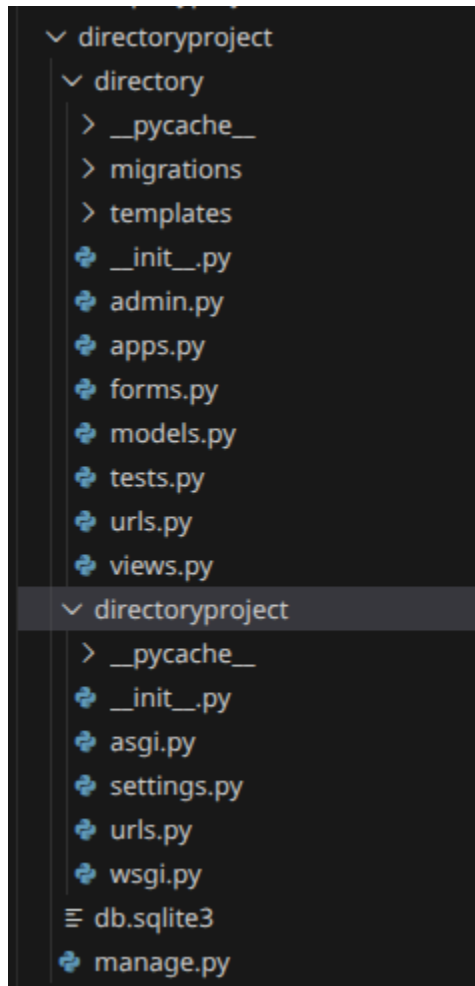


Lab 10 - Class Exercises

Prasanna - 230905506 - A2

- 1) Design a web site using Django, which is a website directory – A site containing links to other websites. A web page has different categories.
- A category table has a name, number of visits, and number of likes.
 - A page table refers to a category, has a title, URL, and many views.
- Design a form that populates the above database and displays it.

a) Directory Structure



b) Codes

directoryproject/directoryproject/settings.py

```
"""
Django settings for directoryproject project.

Generated by 'django-admin startproject' using Django 5.2.12.

For more information on this file, see
https://docs.djangoproject.com/en/5.2/topics/settings/

For the full list of settings and their values, see
https://docs.djangoproject.com/en/5.2/ref/settings/
"""

from pathlib import Path

# Build paths inside the project like this: BASE_DIR / 'subdir'.
BASE_DIR = Path(__file__).resolve().parent.parent

# Quick-start development settings - unsuitable for production
# See https://docs.djangoproject.com/en/5.2/howto/deployment/checklist/

# SECURITY WARNING: keep the secret key used in production secret!
SECRET_KEY =
'django-insecure-c#f+-rv3$6#085ps95o#q_wgjpclgkpy#iyutt_oc@va)sr!a'

# SECURITY WARNING: don't run with debug turned on in production!
DEBUG = True

ALLOWED_HOSTS = []

# Application definition

INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
```

```

    'directory',    # ADD THIS
]

MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
    'django.middleware.common.CommonMiddleware',
    'django.middleware.csrf.CsrfViewMiddleware',
    'django.contrib.auth.middleware.AuthenticationMiddleware',
    'django.contrib.messages.middleware.MessageMiddleware',
    'django.middleware.clickjacking.XFrameOptionsMiddleware',
]

ROOT_URLCONF = 'directoryproject.urls'

TEMPLATES = [
    {
        'BACKEND': 'django.template.backends.django.DjangoTemplates',
        'DIRS': [],
        'APP_DIRS': True,
        'OPTIONS': {
            'context_processors': [
                'django.template.context_processors.request',
                'django.contrib.auth.context_processors.auth',
                'django.contrib.messages.context_processors.messages',
            ],
        },
    },
]

WSGI_APPLICATION = 'directoryproject.wsgi.application'

# Database
# https://docs.djangoproject.com/en/5.2/ref/settings/#databases

DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.sqlite3',

```

```
        'NAME': BASE_DIR / 'db.sqlite3',
    }
}

# Password validation
#
https://docs.djangoproject.com/en/5.2/ref/settings/#auth-password-validators

AUTH_PASSWORD_VALIDATORS = [
    {
        'NAME':
        'django.contrib.auth.password_validation.UserAttributeSimilarityValidator'
    },
    {
        'NAME':
        'django.contrib.auth.password_validation.MinimumLengthValidator',
    },
    {
        'NAME':
        'django.contrib.auth.password_validation.CommonPasswordValidator',
    },
    {
        'NAME':
        'django.contrib.auth.password_validation.NumericPasswordValidator',
    },
]

# Internationalization
# https://docs.djangoproject.com/en/5.2/topics/i18n/

LANGUAGE_CODE = 'en-us'

TIME_ZONE = 'UTC'

USE_I18N = True
```

```
USE_TZ = True

# Static files (CSS, JavaScript, Images)
# https://docs.djangoproject.com/en/5.2/howto/static-files/

STATIC_URL = 'static/'

# Default primary key field type
# https://docs.djangoproject.com/en/5.2/ref/settings/#default-auto-field

DEFAULT_AUTO_FIELD = 'django.db.models.BigAutoField'
```

directoryproject/directory/urls.py

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.home, name='home'),
    path('add-category/', views.add_category, name='add_category'),
    path('add-page/', views.add_page, name='add_page'),
]
```

directoryproject/directoryproject/urls.py

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('directory.urls')),
]
```

directoryproject/directory/forms.py

```
from django import forms
from .models import Category, Page
```

```
class CategoryForm(forms.ModelForm):
    class Meta:
        model = Category
        fields = ['name', 'visits', 'likes']

class PageForm(forms.ModelForm):
    class Meta:
        model = Page
        fields = ['category', 'title', 'url', 'views']
```

directoryproject/directory/templates/home.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Website Directory</title>

    <style>
        body {
            font-family: 'Segoe UI', Arial;
            background: linear-gradient(to right, #eef2f7, #e0e7ff);
            margin: 0;
        }

        header {
            background: linear-gradient(135deg, #1e3a8a, #2563eb);
            color: white;
            padding: 20px;
            text-align: center;
            font-size: 24px;
            font-weight: bold;
            letter-spacing: 1px;
        }

        .container {
            width: 85%;
            margin: 30px auto;
        }
    </style>
</head>
```

```
.top-actions {
    text-align: center;
    margin-bottom: 25px;
}

.btn {
    display: inline-block;
    margin: 5px;
    padding: 10px 16px;
    background: #2563eb;
    color: white;
    border-radius: 25px;
    text-decoration: none;
    font-size: 14px;
    transition: 0.3s;
}

.btn:hover {
    background: #1d4ed8;
    transform: scale(1.05);
}

.category-card {
    background: white;
    border-radius: 15px;
    padding: 20px;
    margin-bottom: 25px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.08);
    transition: 0.3s;
    position: relative;
    overflow: hidden;
}

.category-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 15px 35px rgba(0,0,0,0.15);
}

/* gradient strip */
```

```
.category-card::before {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  width: 100%;
  height: 6px;
  background: linear-gradient(to right, #2563eb, #7c3aed);
}

.category-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.category-name {
  font-size: 20px;
  font-weight: 600;
  color: #1e293b;
}

.stats {
  font-size: 13px;
}

.badge {
  background: #f1f5f9;
  padding: 5px 10px;
  border-radius: 20px;
  margin-left: 5px;
  font-weight: 500;
}

.pages {
  margin-top: 15px;
}

.page-item {
  background: #f8fafc;
}
```



```
        padding: 12px;
        border-radius: 8px;
        margin-bottom: 10px;
        transition: 0.2s;
    }

    .page-item:hover {
        background: #e0e7ff;
        transform: scale(1.01);
    }

    .page-title {
        font-weight: 600;
        color: #1e293b;
    }

    .page-link {
        font-size: 13px;
        color: #2563eb;
        text-decoration: none;
    }

    .page-link:hover {
        text-decoration: underline;
    }

    .page-views {
        font-size: 12px;
        color: #64748b;
    }

    .empty {
        color: #888;
        font-size: 14px;
    }

</style>
</head>

<body>
```

```

<header>Website Directory</header>

<div class="container">

    <div class="top-actions">
        <a href="/add-category/" class="btn">+ Add Category</a>
        <a href="/add-page/" class="btn">+ Add Page</a>
    </div>

    {% for category in categories %}

        <div class="category-card">

            <!-- Category Header -->
            <div class="category-header">
                <div class="category-name">📁 {{ category.name }}</div>

                <div class="stats">
                    <span class="badge">👁️ {{ category.visits }}</span>
                    <span class="badge">❤️ {{ category.likes }}</span>
                </div>
            </div>

            <!-- Pages -->
            <div class="pages">

                {% for page in category.page_set.all %}
                    <div class="page-item">
                        <div class="page-title">{{ page.title }}</div>
                        <div class="page-views">Views: {{ page.views
}}</div>

                        <a href="{{ page.url }}" target="_blank"
class="page-link">
                            {{ page.url }}
                        </a>
                    </div>
                {% empty %}
                    <p class="empty">No pages in this category.</p>
                {% endfor %}
            </div>
        </div>
    {% endfor %}

```

```

        </div>

    </div>

    {% endfor %}

</div>

</body>
</html>

```

directoryproject/directory/templates/add_page.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Add Page</title>

    <style>
        body {
            font-family: Arial;
            background: #f1f5f9;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .box {
            background: white;
            padding: 30px;
            width: 400px;
            border-radius: 10px;
            box-shadow: 0 4px 12px rgba(0,0,0,0.1);
        }

        h2 {
            text-align: center;
            margin-bottom: 20px;

```

```

    }

    input, select {
        width: 100%;
        padding: 8px;
        margin: 10px 0;
        border-radius: 5px;
        border: 1px solid #ccc;
    }

    button {
        width: 100%;
        padding: 10px;
        background: #16a34a;
        color: white;
        border: none;
        border-radius: 5px;
    }

    button:hover {
        background: #15803d;
    }

    a {
        display: block;
        text-align: center;
        margin-top: 15px;
    }
</style>
</head>

<body>

<div class="box">

    <h2>Add Page</h2>

    <form method="POST">
        {% csrf_token %}
        {{ form.as_p }}

```

```
        <button type="submit">Save</button>
    </form>

    <a href="/">← Back</a>

</div>

</body>
</html>
```

directoryproject/directory/templates/add_category.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Add Category</title>

    <style>
        body {
            font-family: Arial;
            background: #f1f5f9;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .box {
            background: white;
            padding: 30px;
            width: 350px;
            border-radius: 10px;
            box-shadow: 0 4px 12px rgba(0,0,0,0.1);
        }

        h2 {
            text-align: center;
            margin-bottom: 20px;
        }

        input {
```

```

        width: 100%;
        padding: 8px;
        margin: 10px 0;
        border-radius: 5px;
        border: 1px solid #ccc;
    }

    button {
        width: 100%;
        padding: 10px;
        background: #2563eb;
        color: white;
        border: none;
        border-radius: 5px;
    }

    button:hover {
        background: #1d4ed8;
    }

    a {
        display: block;
        text-align: center;
        margin-top: 15px;
    }
</style>
</head>

<body>

<div class="box">

    <h2>Add Category</h2>

    <form method="POST">
        {% csrf_token %}
        {{ form.as_p }}
        <button type="submit">Save</button>
    </form>

```

```
<a href="/">- Back</a>

</div>

</body>
</html>
```

directoryproject/directory/views.py

```
from django.shortcuts import render, redirect
from .models import Category, Page
from .forms import CategoryForm, PageForm

def home(request):

    categories = Category.objects.all()
    pages = Page.objects.all()

    return render(request, 'home.html', {
        'categories': categories,
        'pages': pages
    })

def add_category(request):

    if request.method == 'POST':
        form = CategoryForm(request.POST)

        if form.is_valid():
            form.save()
            return redirect('home')

    else:
        form = CategoryForm()

    return render(request, 'add_category.html', {'form': form})
```

```
def add_page(request):

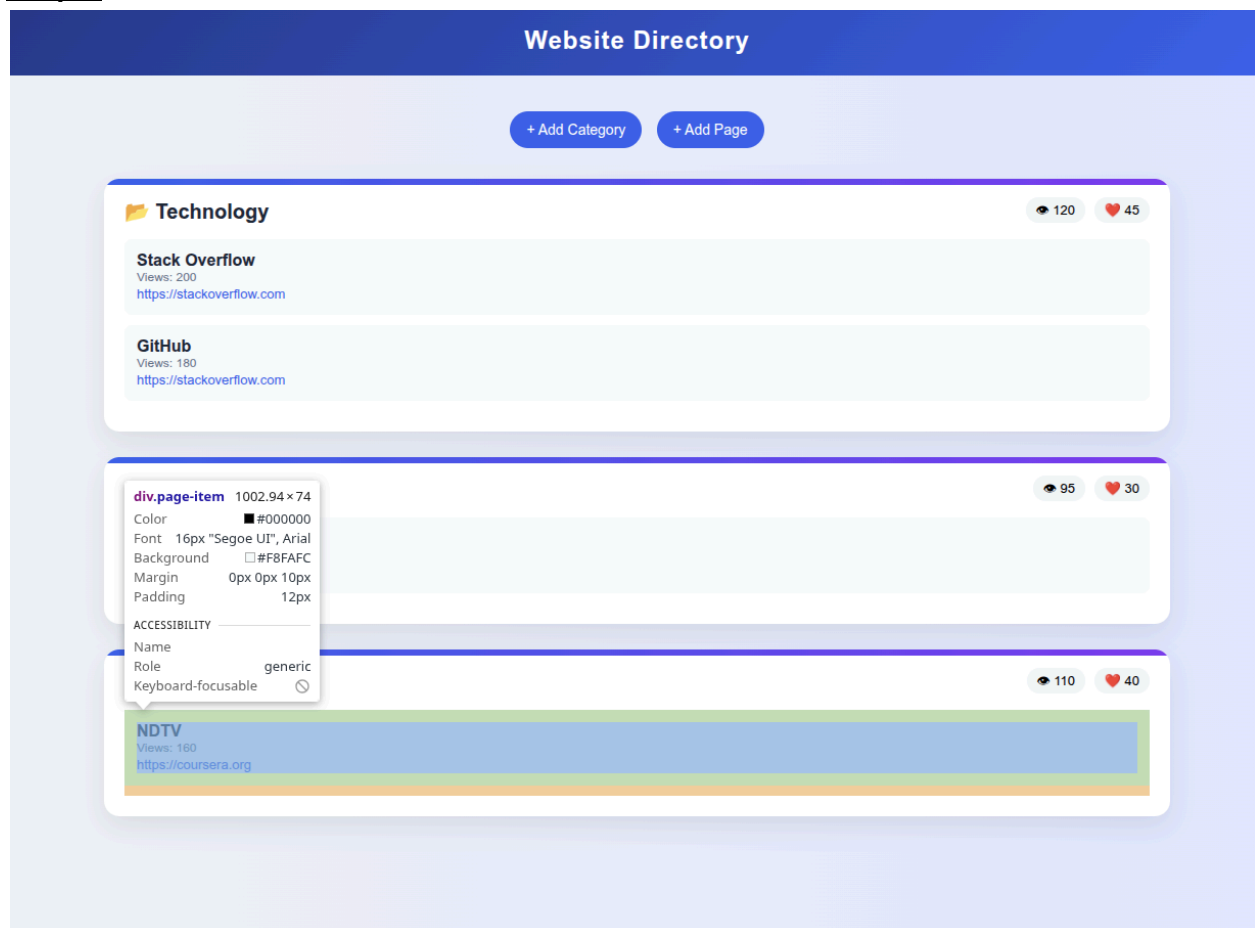
    if request.method == 'POST':
        form = PageForm(request.POST)

        if form.is_valid():
            form.save()
            return redirect('home')

    else:
        form = PageForm()

    return render(request, 'add_page.html', {'form': form})
```

Output



Add Category

Name:

Visits:

Likes:

[Save](#)

[← Back](#)

Add Page

Category:

-----▼

Title:

Url:

Views:

[Save](#)

[← Back](#)

2) Consider the following tables:

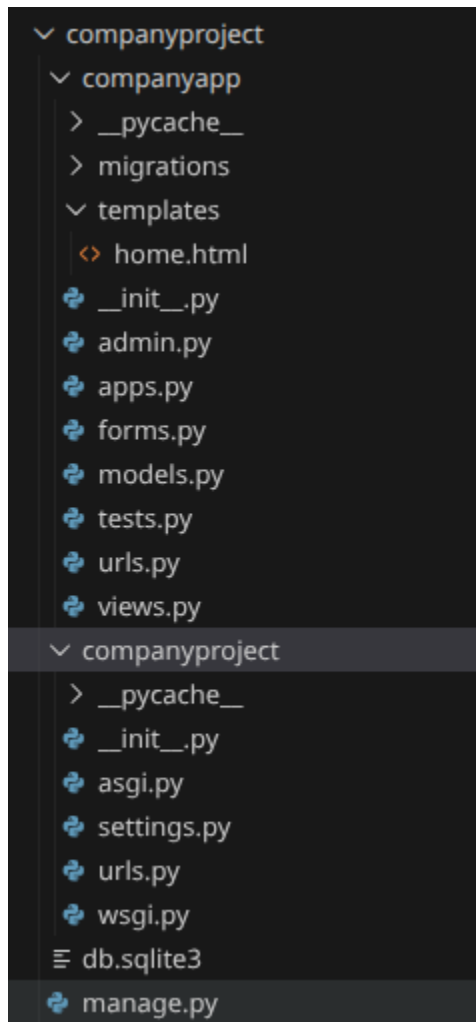
WORKS(person-name,Company-name,Salary)

LIVES(Person_name, Street, City)

Assume Table data suitably. Design a Django webpage and include an option to

insert data into WORKS table by accepting data from the user using TextBoxes. Also, include an option to retrieve the names of people who work for a particular company along with the cities they live in (particular company name must be accepted from the user).

a) Directory Structure



b) Codes

companyproject/companyproject/settings.py

```
"""
Django settings for companyproject project.

Generated by 'django-admin startproject' using Django 5.2.12.
```

```
For more information on this file, see
https://docs.djangoproject.com/en/5.2/topics/settings/

For the full list of settings and their values, see
https://docs.djangoproject.com/en/5.2/ref/settings/
"""

from pathlib import Path

# Build paths inside the project like this: BASE_DIR / 'subdir'.
BASE_DIR = Path(__file__).resolve().parent.parent

# Quick-start development settings - unsuitable for production
# See https://docs.djangoproject.com/en/5.2/howto/deployment/checklist/

# SECURITY WARNING: keep the secret key used in production secret!
SECRET_KEY =
'django-insecure-%gvxw^(9s+j)h93n+6!x=aw8l1&6iu!5lmug0yw-%sth+l^#l!'

# SECURITY WARNING: don't run with debug turned on in production!
DEBUG = True

ALLOWED_HOSTS = []

# Application definition

INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'companyapp',
]

MIDDLEWARE = [
```

```

'django.middleware.security.SecurityMiddleware',
'django.contrib.sessions.middleware.SessionMiddleware',
'django.middleware.common.CommonMiddleware',
'django.middleware.csrf.CsrfViewMiddleware',
'django.contrib.auth.middleware.AuthenticationMiddleware',
'django.contrib.messages.middleware.MessageMiddleware',
'django.middleware.clickjacking.XFrameOptionsMiddleware',
]

ROOT_URLCONF = 'companyproject.urls'

TEMPLATES = [
    {
        'BACKEND': 'django.template.backends.django.DjangoTemplates',
        'DIRS': [],
        'APP_DIRS': True,
        'OPTIONS': {
            'context_processors': [
                'django.template.context_processors.request',
                'django.contrib.auth.context_processors.auth',
                'django.contrib.messages.context_processors.messages',
            ],
        },
    ],
]

WSGI_APPLICATION = 'companyproject.wsgi.application'

# Database
# https://docs.djangoproject.com/en/5.2/ref/settings/#databases

DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.sqlite3',
        'NAME': BASE_DIR / 'db.sqlite3',
    }
}

```

```
# Password validation
#
https://docs.djangoproject.com/en/5.2/ref/settings/#auth-password-validators

AUTH_PASSWORD_VALIDATORS = [
    {
        'NAME':
'django.contrib.auth.password_validation.UserAttributeSimilarityValidator'
    },
    {
        'NAME':
'django.contrib.auth.password_validation.MinimumLengthValidator',
    },
    {
        'NAME':
'django.contrib.auth.password_validation.CommonPasswordValidator',
    },
    {
        'NAME':
'django.contrib.auth.password_validation.NumericPasswordValidator',
    },
]

# Internationalization
# https://docs.djangoproject.com/en/5.2/topics/i18n/

LANGUAGE_CODE = 'en-us'

TIME_ZONE = 'UTC'

USE_I18N = True

USE_TZ = True

# Static files (CSS, JavaScript, Images)
# https://docs.djangoproject.com/en/5.2/howto/static-files/
```

```
STATIC_URL = 'static/'

# Default primary key field type
# https://docs.djangoproject.com/en/5.2/ref/settings/#default-auto-field

DEFAULT_AUTO_FIELD = 'django.db.models.BigAutoField'
```

companyproject/companyapp/urls.py

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.home, name='home'),
]
```

companyproject/companyproject/urls.py

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('companyapp.urls')),
]
```

companyproject/companyapp/views.py

```
from django.shortcuts import render
from .models import Works, Lives
from .forms import CombinedForm, SearchForm

def home(request):

    form = CombinedForm()
    search_form = SearchForm()
    results = []
```

```

# INSERT DATA
if request.method == 'POST' and 'save' in request.POST:
    form = CombinedForm(request.POST)
    if form.is_valid():
        name = form.cleaned_data['person_name']
        company = form.cleaned_data['company_name']
        salary = form.cleaned_data['salary']
        street = form.cleaned_data['street']
        city = form.cleaned_data['city']

        # Save in WORKS
        Works.objects.create(
            person_name=name,
            company_name=company,
            salary=salary
        )

        # Save in LIVES
        Lives.objects.create(
            person_name=name,
            street=street,
            city=city
        )

# SEARCH
if request.method == 'POST' and 'search' in request.POST:
    search_form = SearchForm(request.POST)
    if search_form.is_valid():
        company = search_form.cleaned_data['company_name']

        works = Works.objects.filter(company_name__iexact=company)

        for w in works:
            lives =
Lives.objects.filter(person_name__iexact=w.person_name).first()
            if lives:
                results.append({
                    'name': w.person_name,
                    'city': lives.city

```

```
    })

    return render(request, 'home.html', {
        'form': form,
        'search_form': search_form,
        'results': results
    })
```

companyproject/companyapp/templates/home.html

z

Add Employee Details

Person name:

Company name:

Salary:

Street:

City:

Save Data

Search by Company

Company name:

meta

Search

Results:

Shubham → Bangalore